

Yi Ding

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Ph.D. Student in Computer Science, Purdue University

Website · Google Scholar · GitHub · LinkedIn

EDUCATION

Purdue University

Ph.D. in Computer Science

Advisor: Dr. Ruqi Zhang

IN, USA

Aug. 2025 – Present

Tianjin University

B.S. in Data Science and Big Data Technology

School of Mathematics · Advisors: Dr. Bing Cao and Dr. Qinghua Hu

Tianjin, China

Aug. 2021 – Jun. 2025

RESEARCH INTERESTS

I study how large language and vision-language models can improve from limited, noisy, or imperfect supervision. My work develops reliable and efficient post-training algorithms for reasoning and safety, with a focus on reinforcement learning, on-policy distillation, and self-correction.

SELECTED PUBLICATIONS

First/co-first-author papers and a collaborative technical report. * Equal contribution.

Reasoning & Self-Improvement

[S1] Does On-Policy Distillation Really Distill? From Noisy Teacher to Self-Improvement

Yi Ding, Ruqi Zhang

Preprint 2026 [PDF] [Project]

Supervision-free self-improvement through on-policy self-adaptation.

#1 on Hugging Face Daily Papers (Sep. 1, 2026)

[S2] Learning Self-Correction in Vision-Language Models via Rollout Augmentation

Yi Ding, Ziliang Qiu, Bolian Li, Ruqi Zhang

ICML 2026 [PDF] [Project]

Learning self-correction by augmenting reinforcement-learning rollouts.

[S3] Sherlock: Self-Correcting Reasoning in Vision-Language Models

Yi Ding, Ruqi Zhang

NeurIPS 2025 [PDF] [Project]

Bootstrapping self-correction and self-improvement from limited supervision.

Multimodal Safety & Alignment

[S4] Rethinking Bottlenecks in Safety Fine-Tuning of Vision Language Models

Yi Ding*, Lijun Li*, Bing Cao, Jing Shao

ICLR 2026 [PDF] [Project]

Teaching vision-language models to reason about safety across multiple images.

[S5] SafeWork-R1: Coevolving Safety and Intelligence under the AI-45° Law

Shanghai AI Lab: Yicheng Bao, ..., Yi Ding, ... (100+ authors)

Technical Report 2025 [PDF]

Jointly improving safety and intelligence in multimodal reasoning.

[S6] ETA: Evaluating Then Aligning Safety of Vision Language Models at Inference Time

Yi Ding, Bolian Li, Ruqi Zhang

ICLR 2025 [PDF] [Project]

Training-free safety alignment for vision-language models at inference time.

ADDITIONAL PUBLICATIONS

* Equal contribution.

- [A1] **Addressing Performance Saturation for LLM RL via Precise Entropy Curve Control**
Bolian Li, Yifan Wang, **Yi Ding**, Anamika Lochab, Ananth Grama, Ruqi Zhang
Preprint 2026 [PDF] [Project]
- [A2] **Position: Modular Safety Guardrails Are Necessary for Foundation-Model-Enabled Robots in the Real World**
Joonkyung Kim, Wenxi Chen, Davood Soleymanzadeh, **Yi Ding**, Xiangbo Gao, Zhengzhong Tu, Ruqi Zhang, Fan Fei, Sushant Veer, Yiwei Lyu, Minghui Zheng, Yan Gu
ICML 2026 (Position Paper) [PDF]
- [A3] **Visual Contextual Attack: Jailbreaking MLLMs with Image-Driven Context Injection**
Ziqi Miao*, **Yi Ding***, Lijun Li, Jing Shao
EMNLP 2025 (Main) [PDF]
- [A4] **Test-Time Dynamic Image Fusion**
Bing Cao, Yinan Xia*, **Yi Ding***, Changqing Zhang, Qinghua Hu
NeurIPS 2024 [PDF]
- [A5] **Predictive Dynamic Fusion**
Bing Cao, Yinan Xia*, **Yi Ding***, Changqing Zhang, Qinghua Hu
ICML 2024 [PDF]

RESEARCH EXPERIENCE

RZ-Lab, Purdue University

Research Assistant

Advisor: Dr. Ruqi Zhang

IN, USA

May 2024 – May 2025

Open Trust Lab, Shanghai AI Laboratory

Research Scientist

Advisor: Dr. Lijun Li

Beijing, China

Dec. 2024 – Mar. 2025

MLDM Lab, Tianjin University

Research Assistant

Advisors: Dr. Bing Cao and Dr. Qinghua Hu

Tianjin, China

Sep. 2023 – Dec. 2024

AWARDS AND RESEARCH SUPPORT

NSF ACCESS Discover Project Award

Project: LLM Post-Training

2025–2026

ACADEMIC SERVICE

Reviewer: ICLR 2025–2027; ICML 2026; NeurIPS 2025, 2026; CIKM 2026; ACL Rolling Review (May 2025).

SKILLS AND LANGUAGES

Technical: Python, MATLAB, $\text{\LaTeX}$; mathematical and statistical analysis.

Languages: Mandarin Chinese (native); English (TOEFL: 102).